DIPLOMA IN IT IN NETWORK DESIGN AND ADMINISTRATION

FRIDAY PRACTICAL

FA3 POE

NETWORKS 511



CTU training solutions | Private Tertiary Education

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End of Assignment ..... Error! Bookmark not defined.

ASSESSMENT RULES AND REGULATIONS

Instructions to Students:

- This project consists of 10 weekly tasks, each with specific deliverables. Students must complete all tasks as outlined in the rubric.
- Students must submit all weekly evidence to GitHub as they progress through the project.
- By 12 May 2026, students must download their entire GitHub repository as a ZIP folder and upload it on Campus Online for assessment.
- Type all documents and reports in Microsoft Word and convert them to PDF before uploading.

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- Use Calibri font, size 11, for all written submissions.
- Complete the Declaration of Authenticity and include it in the submission.
- Use the Harvard referencing method and provide a minimum of five (5) references. Three (3) marks will be deducted from the final total if the work is not properly referenced.
- Include a Table of Contents in your submission. Failure to do so will result in a deduction of two (2) marks from the final total Mark.

You are required to make use of the Harvard Referencing method. Not referencing your work constitutes plagiarism.

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DEFERMENT APPLICATION

Your deferment application will only be considered if submitted:

Within a **5-day period** after the initial assessment was conducted, and
once the following required documentation is uploaded via CampusOnline:

1. Affidavit signed by a Commissioner of Oaths or the South African Police, **clearly stating the reason**.
2. Doctor's note from a registered health professional.
3. Death Certificate of an immediate family member.

To submit your deferment application:

1. Log into CampusOnline.
2. Click on "Current Courses" and navigate to "Click here to apply for an assessment deferment."

Before applying, please ensure that all required documents are ready for upload to complete the process successfully.

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DECLARATION OF AUTHENTICITY AND RULES AND REGULATIONS - THEORY

I _Ronnie Jim Mokhari_____
(FULL NAME)

hereby declare that the contents of this assessment

_**yes**_____

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any **AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.**

Signature: _RJM_____

Date: _24/06/2026_____

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INTRODUCTION

In today's digital world, networks form the backbone of communication and business operations. Understanding how devices such as routers, switches, and PCs interact is essential for anyone pursuing a career in IT and networking. This assignment is designed to provide students with **hands-on experience in designing, configuring, and troubleshooting networks** using Cisco Packet Tracer, a widely used network simulation tool.

Over the course of five weeks, students will progressively build practical skills, starting from basic device connectivity, IP addressing, and subnetting, to more advanced concepts such as VLAN configuration, inter-VLAN routing, and network documentation. Each week includes tasks that require students to **plan, implement, test, and document their networks**, ensuring both technical understanding and professional reporting skills.

By completing this assignment, students will not only gain proficiency in **Cisco IOS configuration and network simulation**, but also develop **problem-solving, logical thinking, and troubleshooting skills**, which are fundamental to careers in networking, systems administration, and IT support.

LEARNING OUTCOMES:

- Configure and manage Cisco routers and switches using IOS
- Implement IP addressing, VLANs, and inter-VLAN routing
- Troubleshoot connectivity using structured Cisco methodology
- Apply network security practices, CDP configuration, and ACLs
- Document network design, configurations, and testing procedures professionally

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QUALIFICATION AND ASSESSMENT DETAILS

Programme Title	DIPLOMA IN IT IN NETWORK DESIGN AND ADMINISTRATION
Module Title	NETWORKS
Module Code	NET511
NQF Level	5
Formative Assessment	FRIDAY PRACTICAL ASSESSMENT (FA3)
Hand out date	19 June 2026
Submission date	26 June 2026
Total Marks	100
Examiner	Linda Dube
Internal Moderator	Onias Sayi
Internal Moderator	Faith Muwishi and Palesa Nkosi

Student Name and Surname	**Jim Mokhari**
Student Number	**20261952**

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Cisco Networking Capstone Practical Assignment

Purpose: Design, Configure, and Manage a Campus Network Infrastructure
Institution: CTU Training Solutions (Scenario)
Programme Level: NQF Level 5
Duration: 5 Weeks (7 April – 12 May 2026)
Submission Method: GitHub Repository & Campus Online ZIP Upload

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1 Project Overview

This capstone project simulates the role of a network engineer responsible for designing, implementing, and managing a secure, scalable, and functional campus network. Students will implement network fundamentals, routing, VLANs, DHCP, IP addressing, and device management, while progressively building a fully connected network environment. Students are expected to document all configurations, validate connectivity, and submit weekly evidence on GitHub.

Learning Outcomes

By completing this project students will demonstrate:

- Configure and manage Cisco routers and switches using IOS
- Implement IP addressing, VLANs, and inter-VLAN routing
- Troubleshoot connectivity using structured Cisco methodology
- Apply network security practices, CDP configuration, and ACLs
- Document network design, configurations, and testing procedures professionally

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2 Required Tools

To complete this assignment, you need:

- **Cisco Packet Tracer**
- **A computer**
- **Internet access**
- **Word (for report)**
- **Screenshot tool**
- **GitHub Account**

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3 GitHub Repository Requirement

Students must create a GitHub repository to submit Task evidence. Repository naming convention:

Once all Evidence has been uploaded on github, download all files and upload all evidence on CampusOnline

CiscoCapstone-StudentName
Example: CiscoCapstone-LindaDube

4 Recommended Repository Structure:

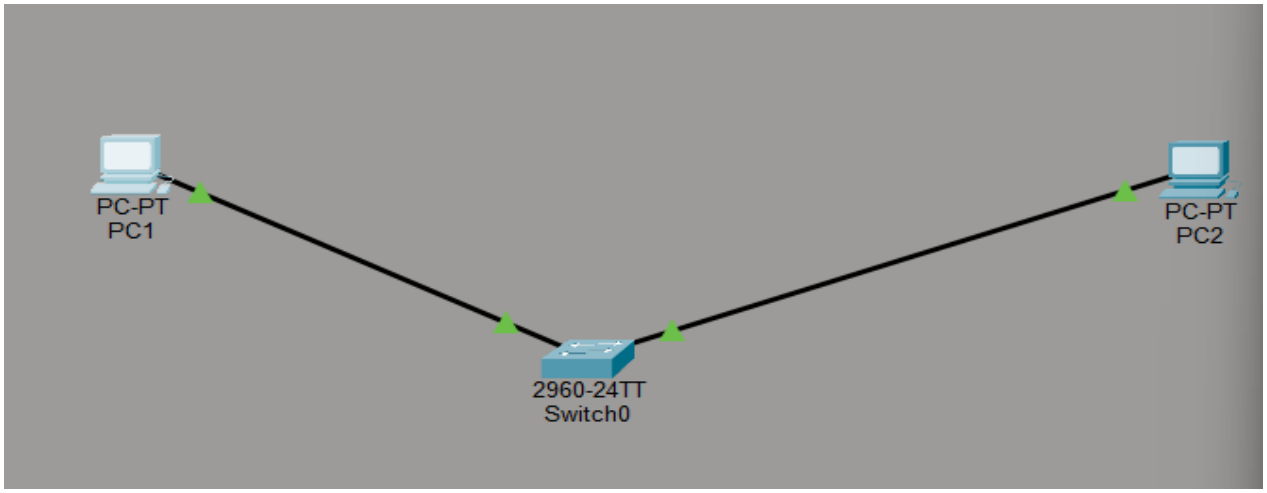
Recommended Repository Structure
Task1_Basic Network
Task2_IP ADDRESING
Task3_VLANS
Task4_Inter-VLAN-Routing
Task5_Final Project

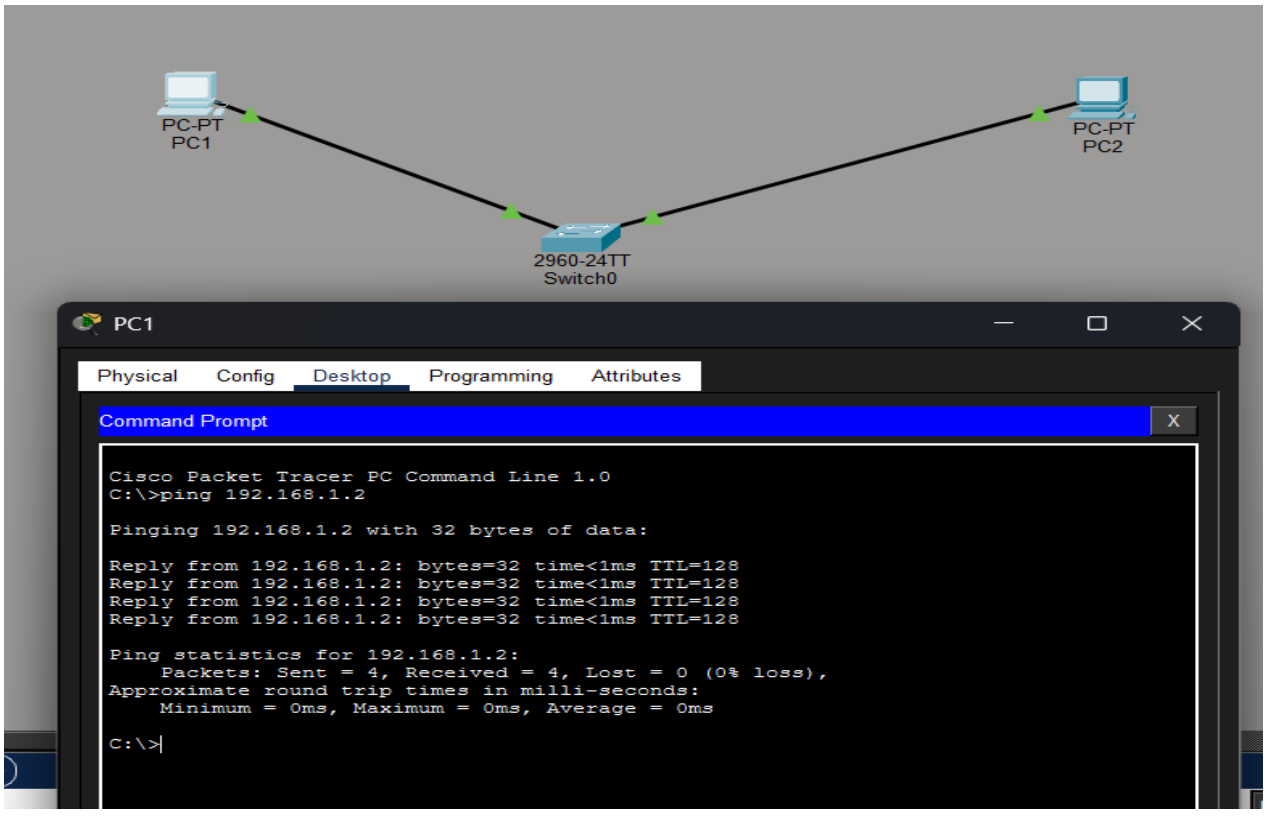
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Assignment: Cisco Packet Tracer Networking Fundamentals
Start Date: 19 June 2026
Submission Date: 26 June 2026
💻 Tool: Cisco Packet Tracer
🎯 **Total Marks: 100**

📅 **Task1: 3–9 April**

Topic: Network Basics & Topology

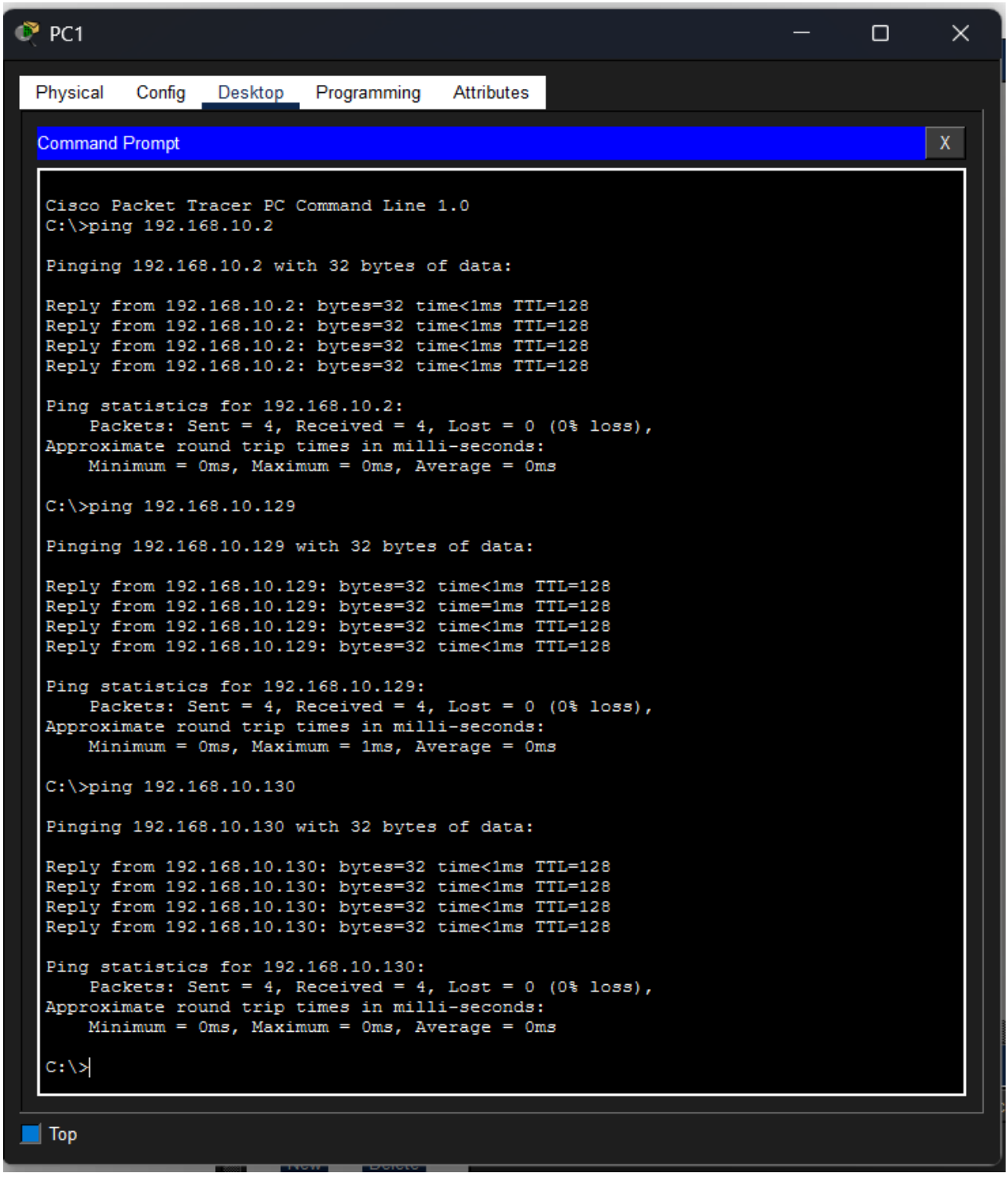




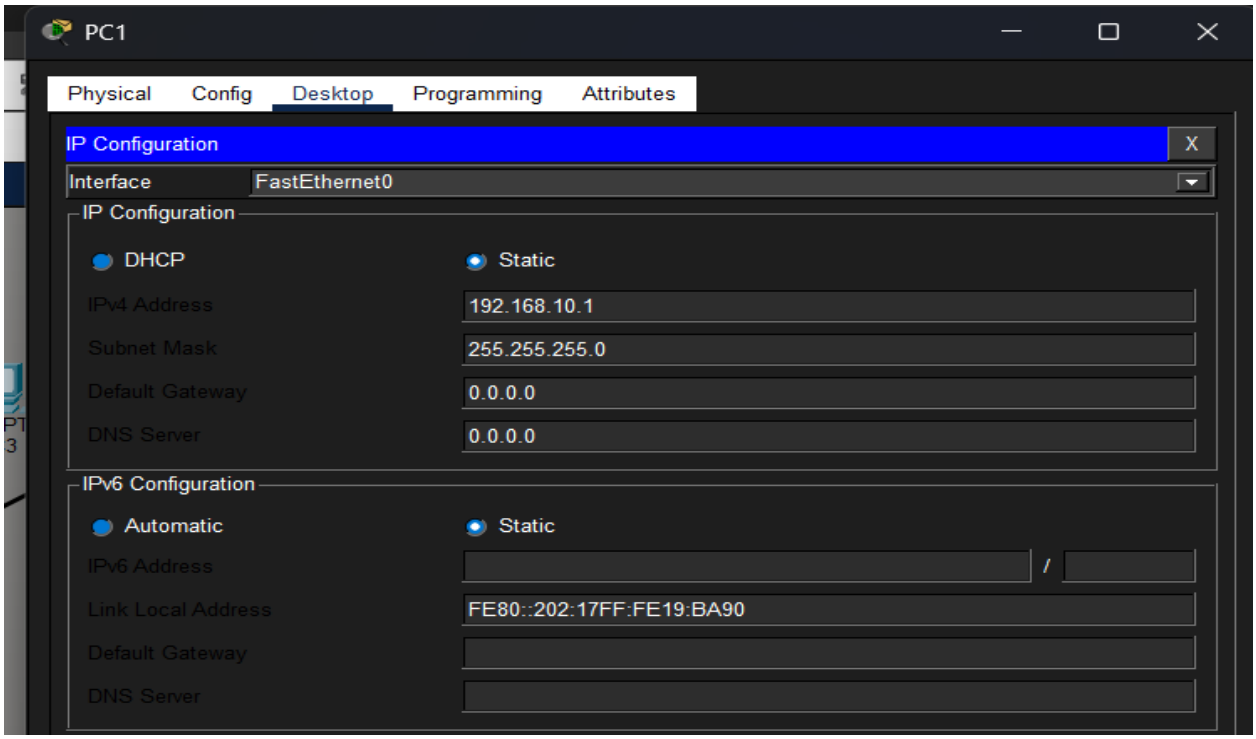
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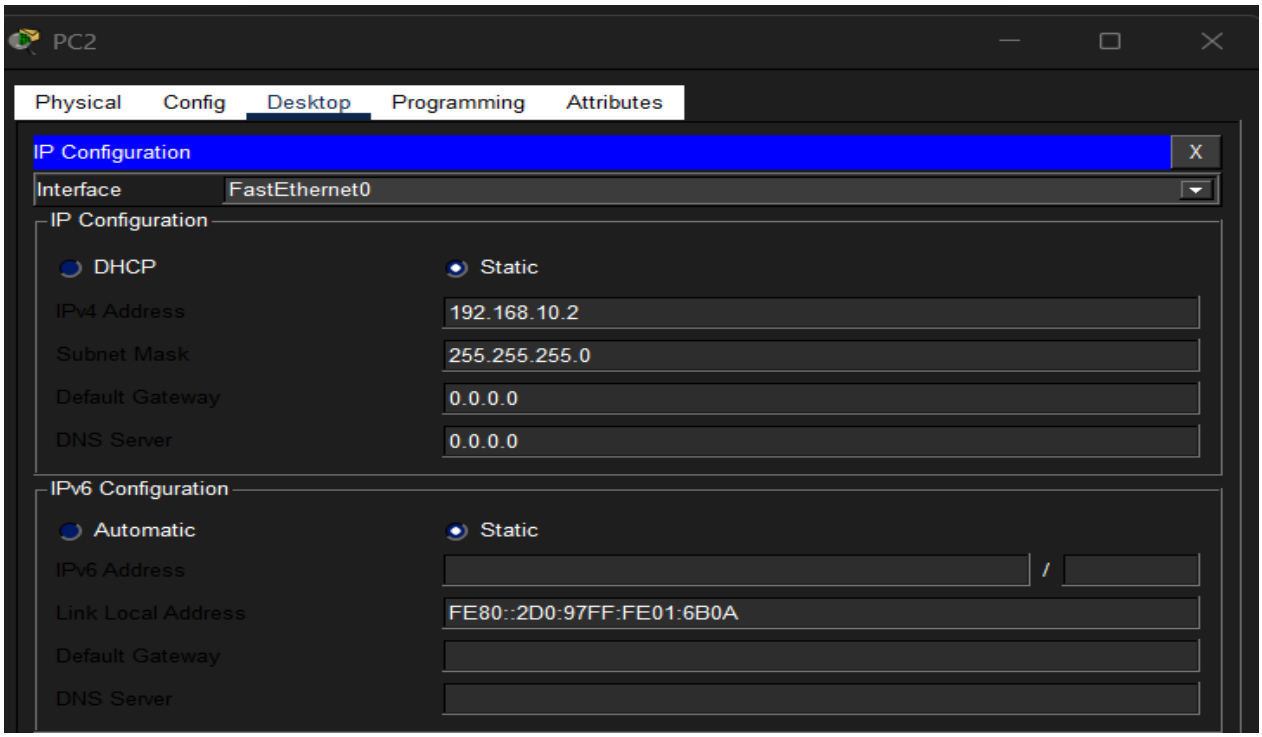
Task2: 10–16 April

- **Topic: IP Addressing & Subnetting**

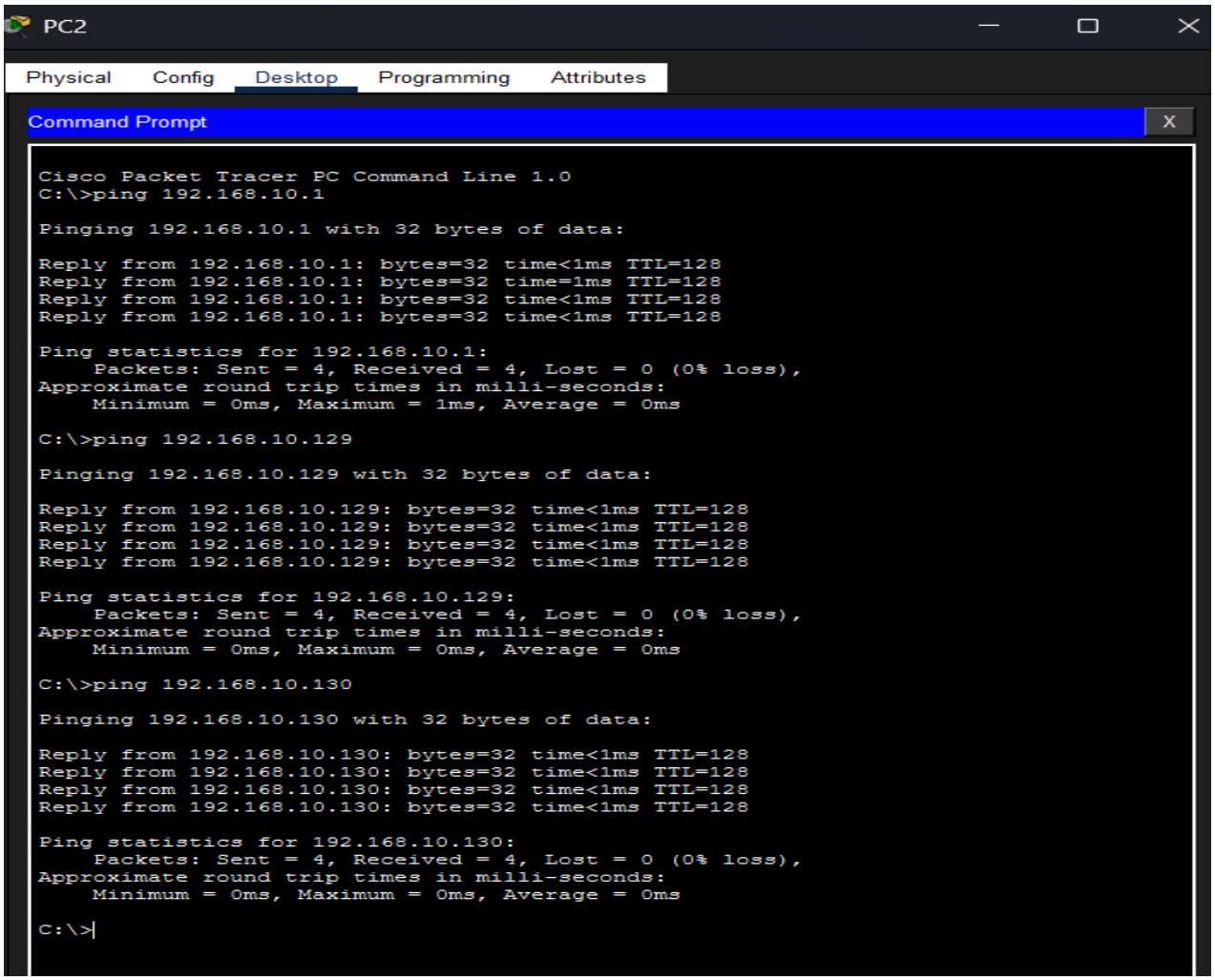


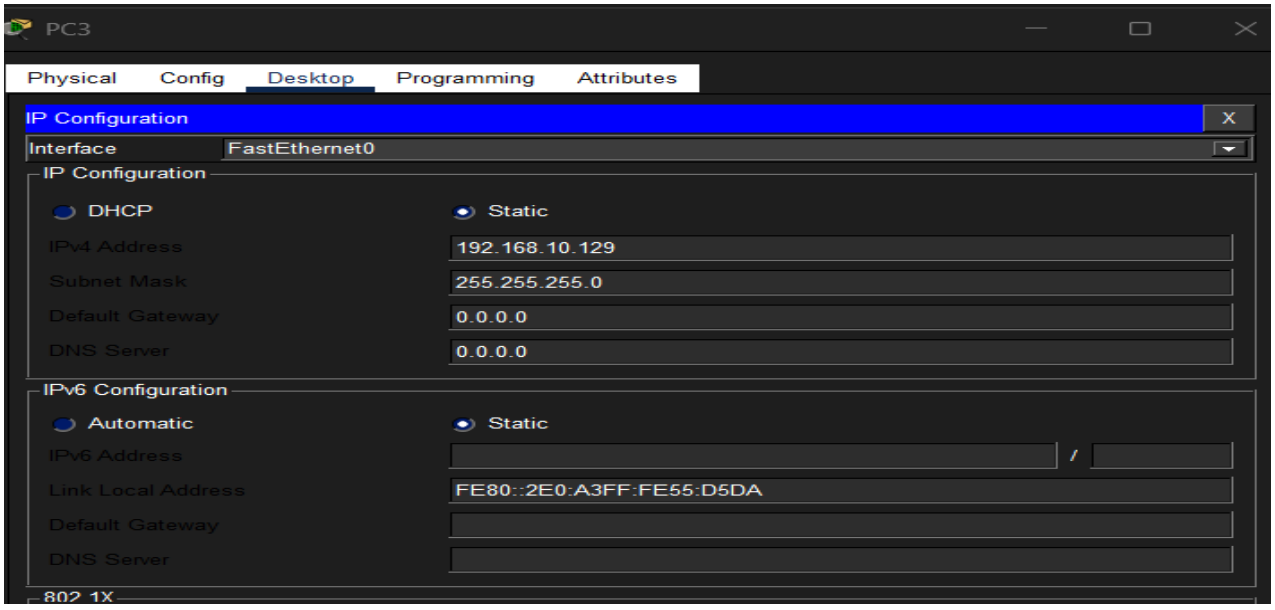
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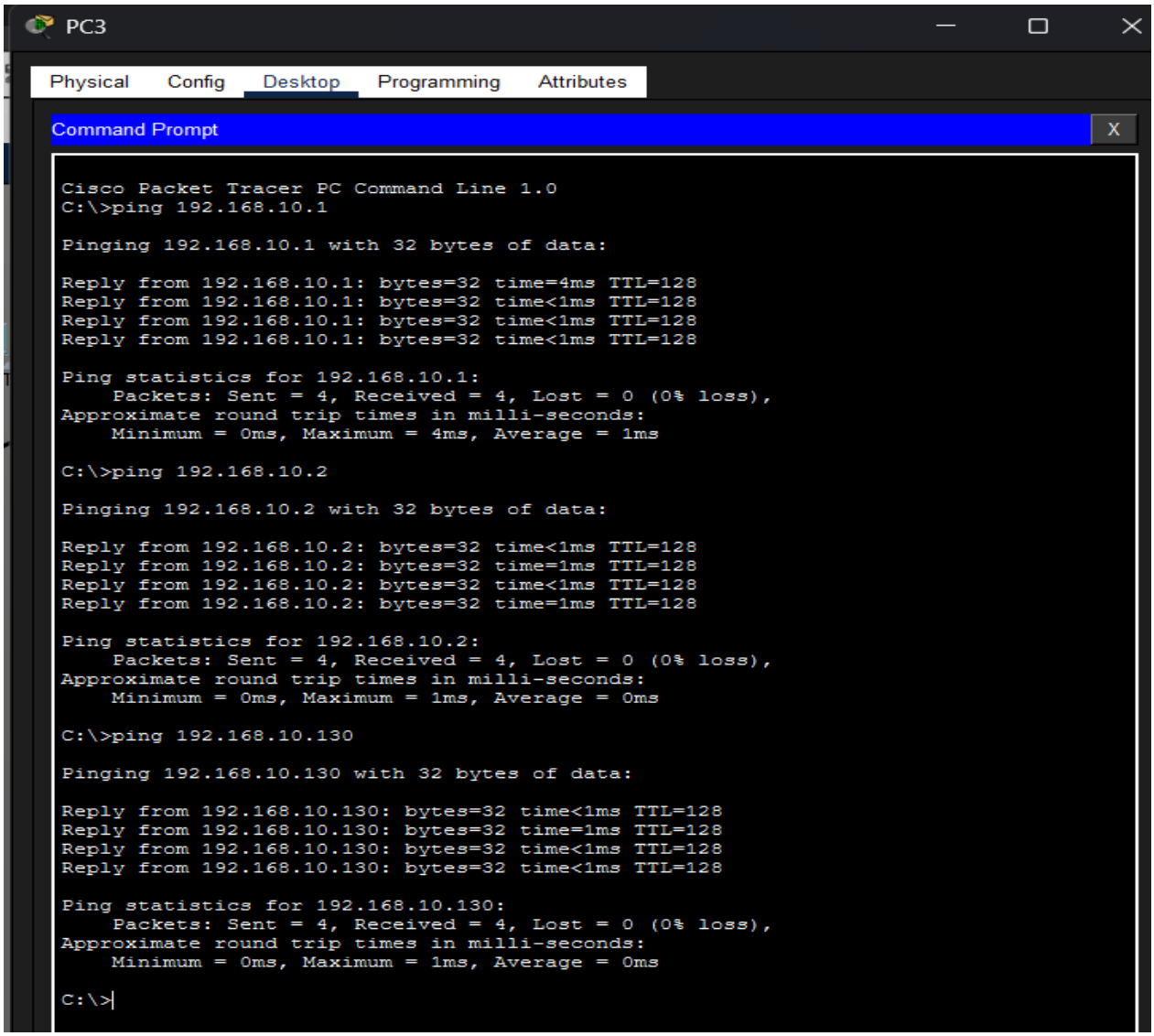


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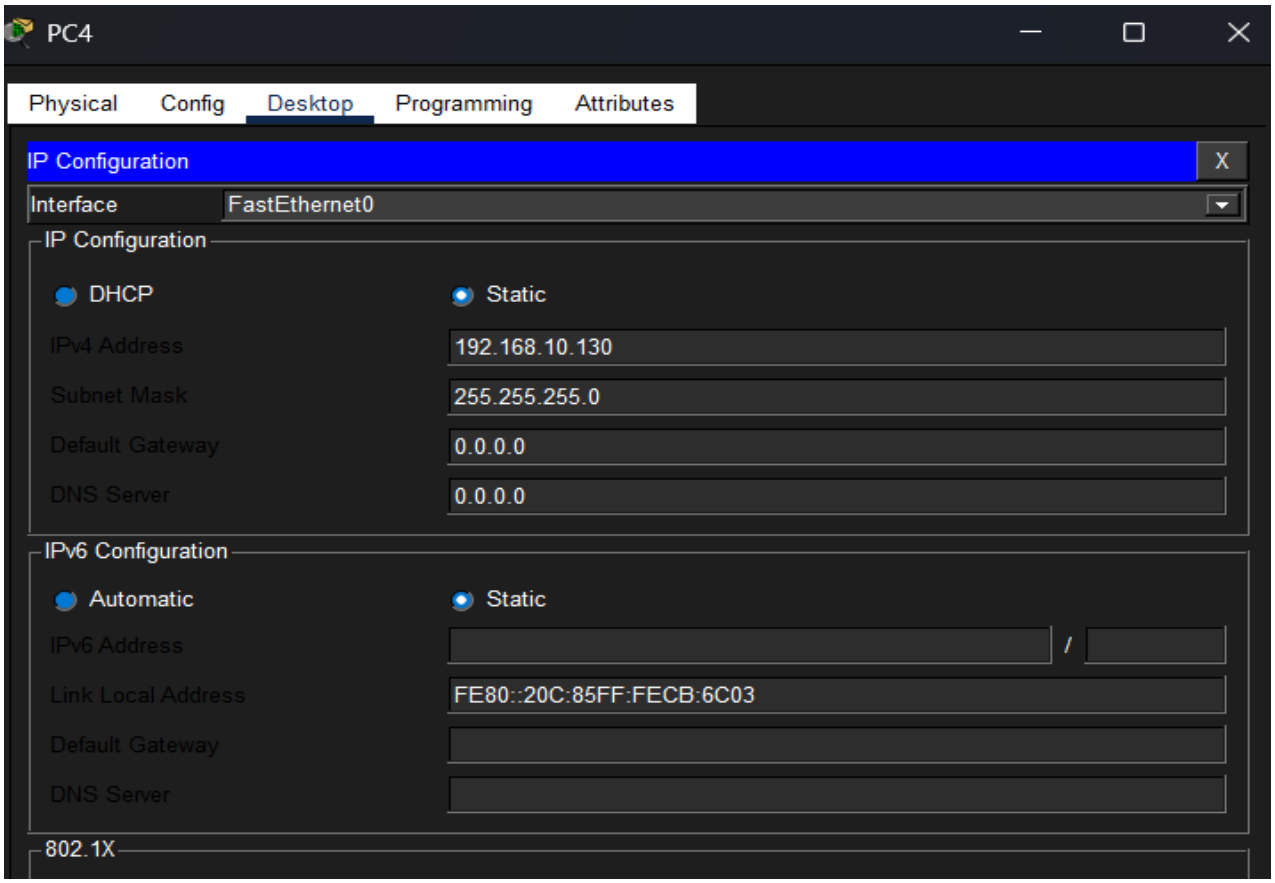




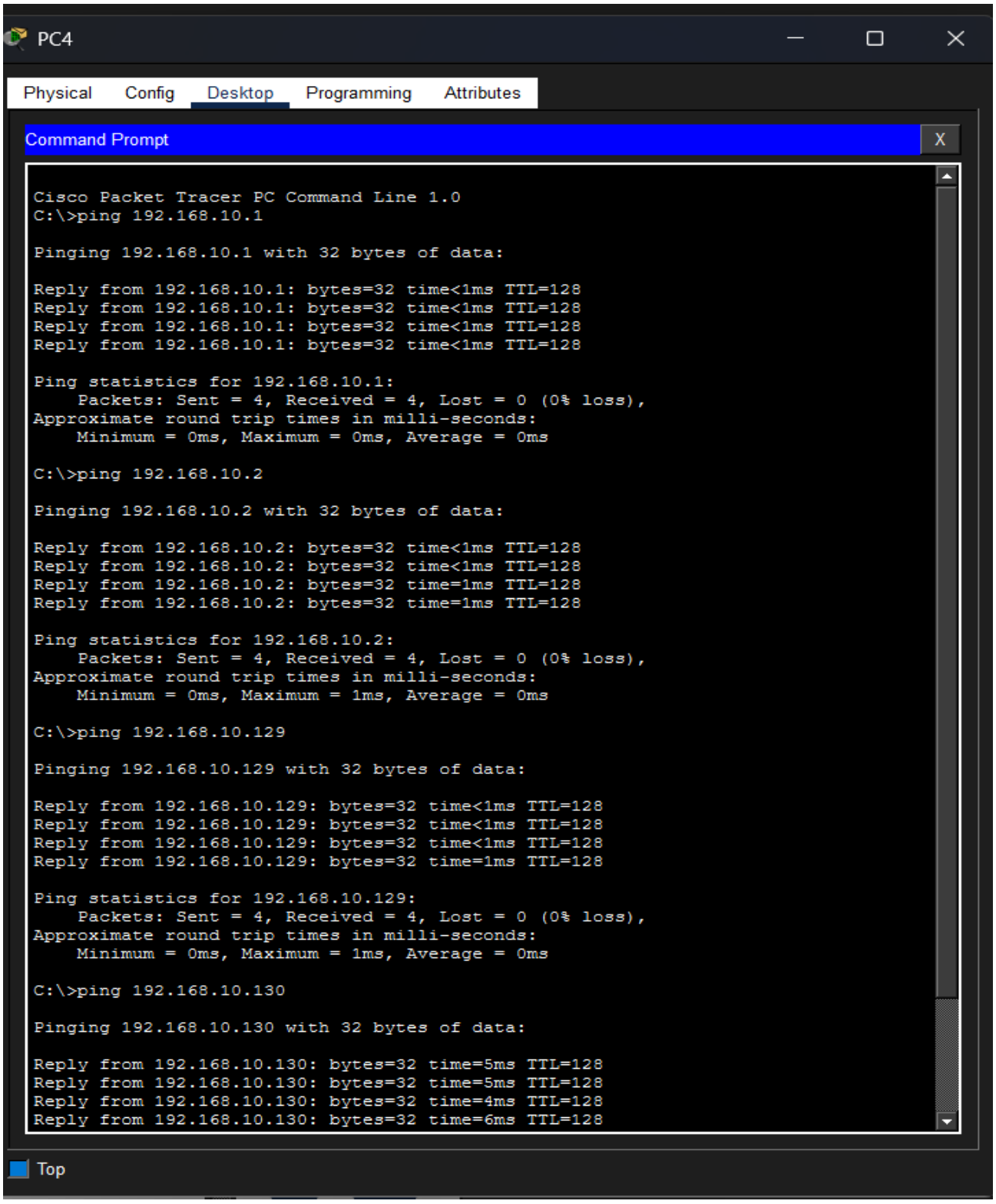
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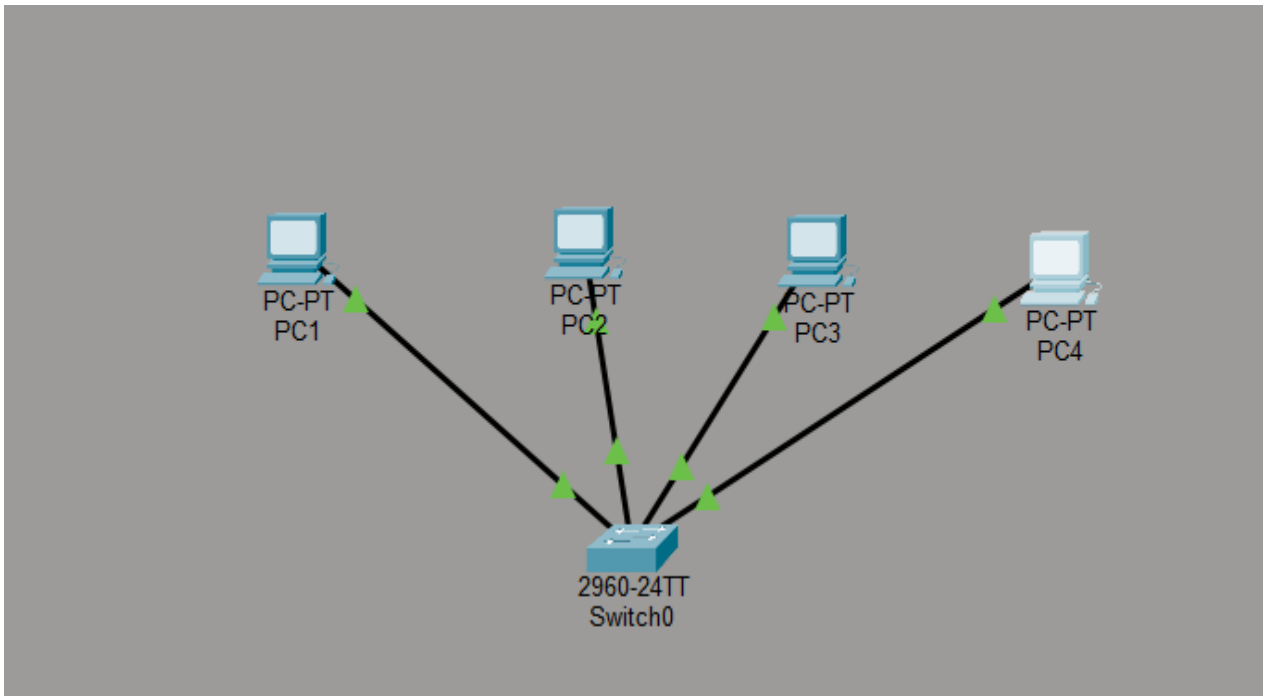
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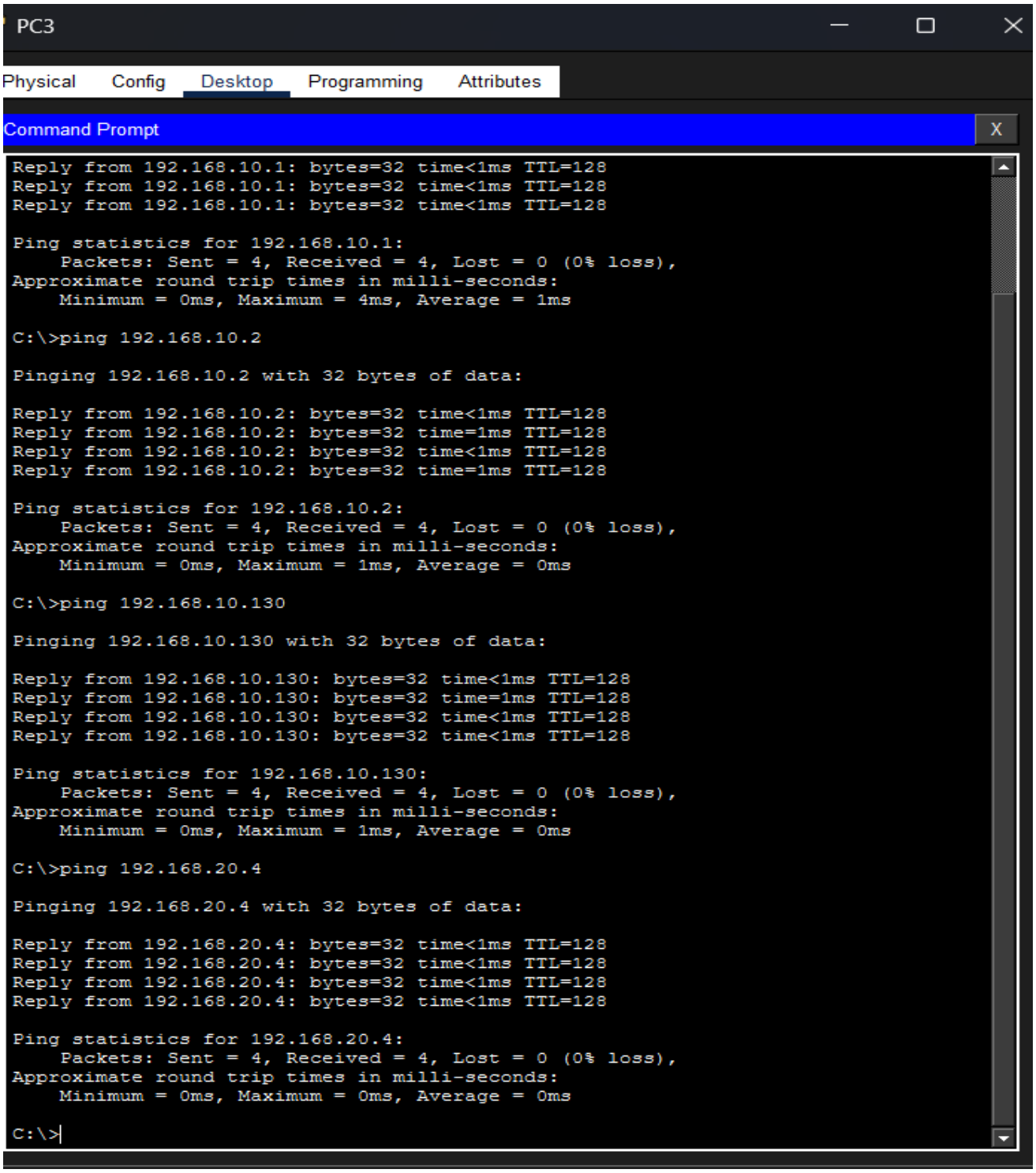
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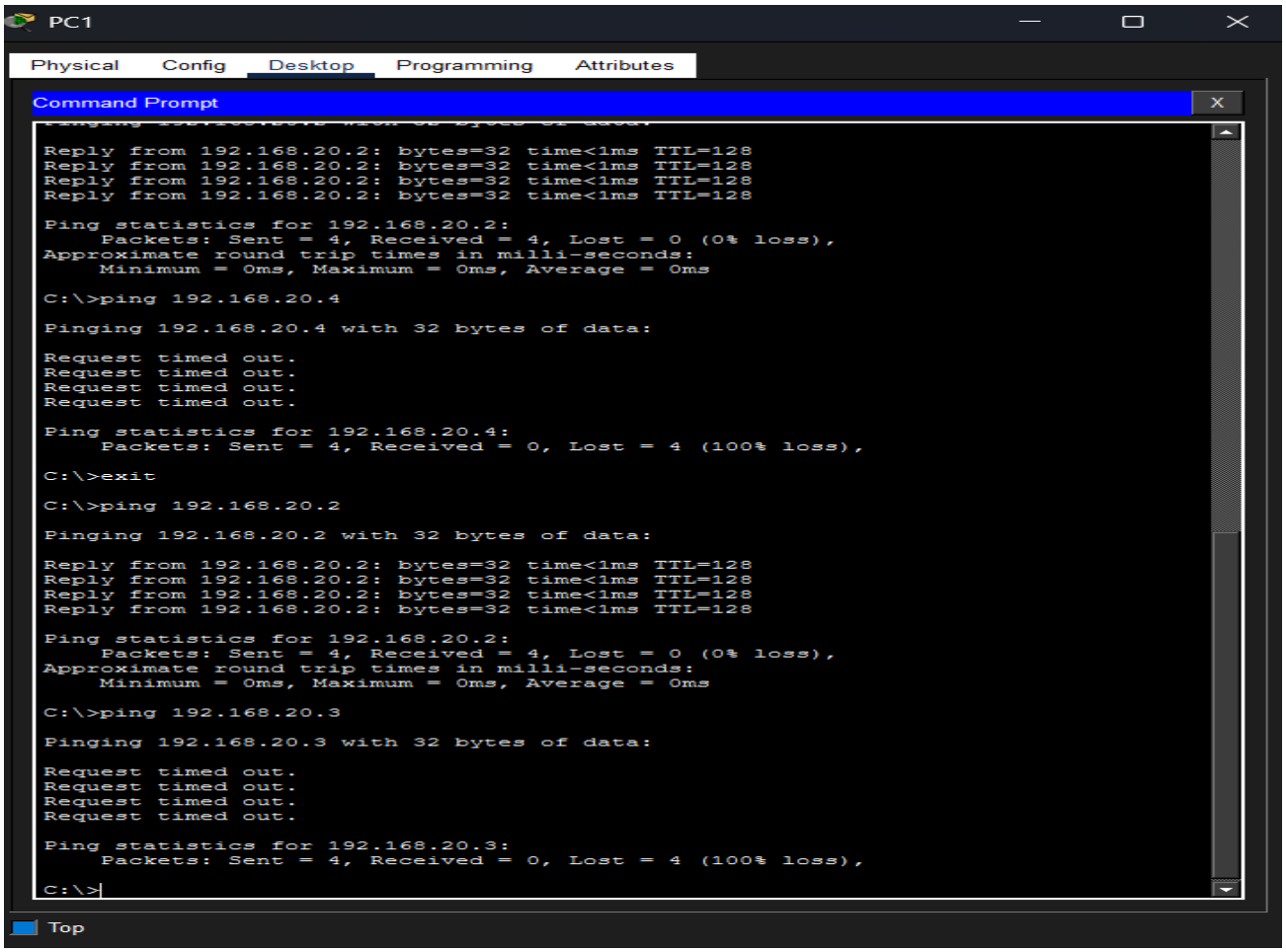
Task3: 17–23 April

- **Topic: VLANs (Basic Segmentation)**

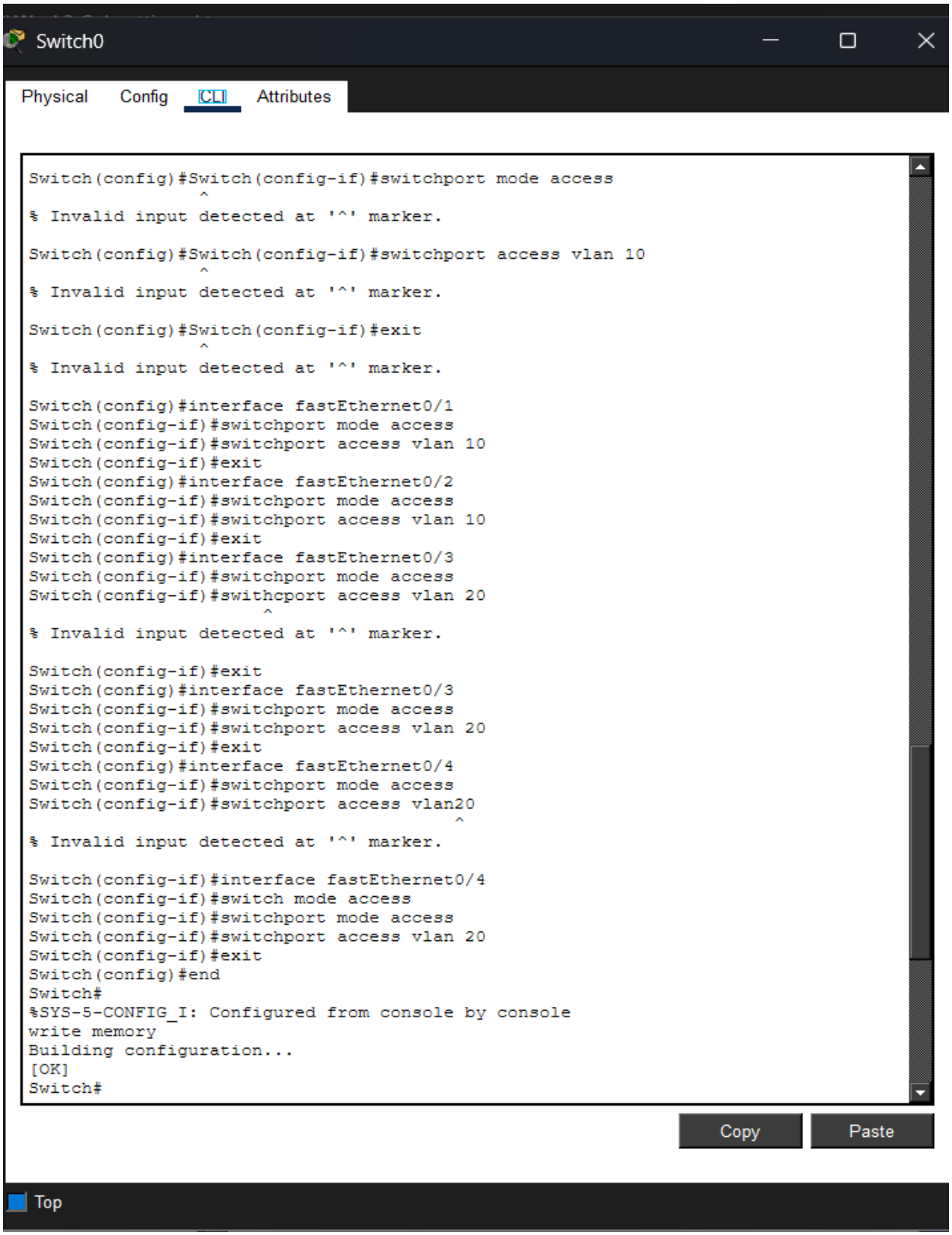
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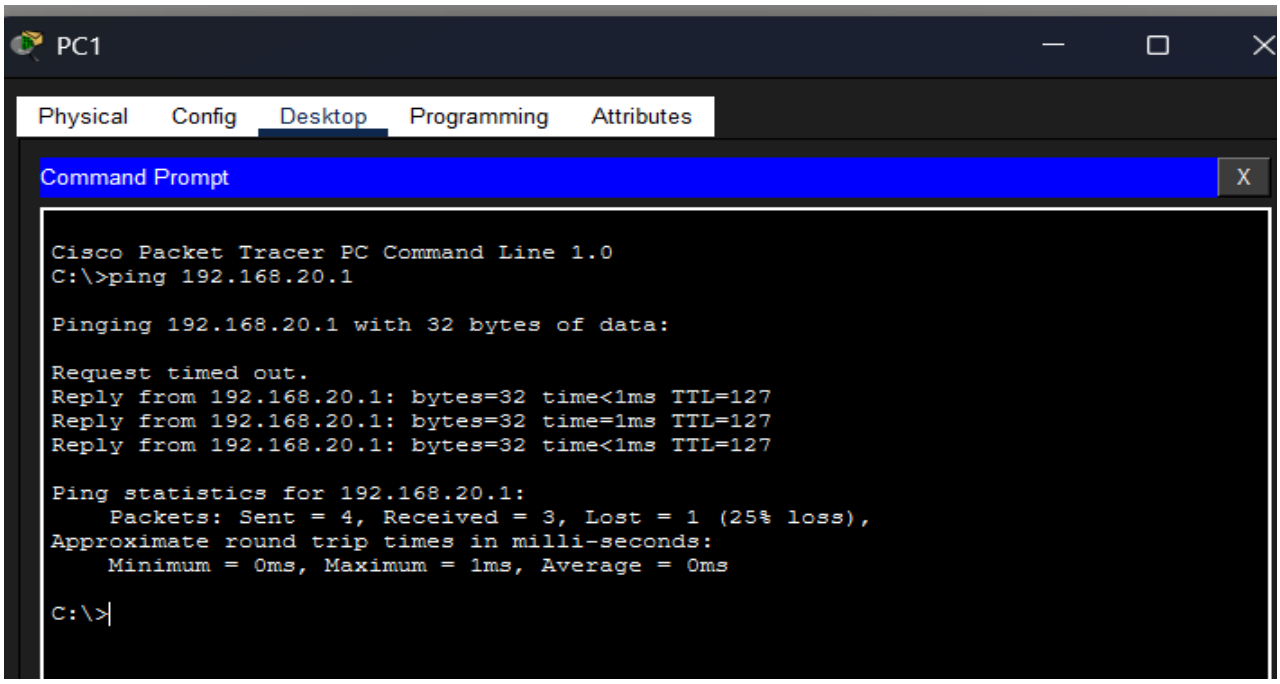
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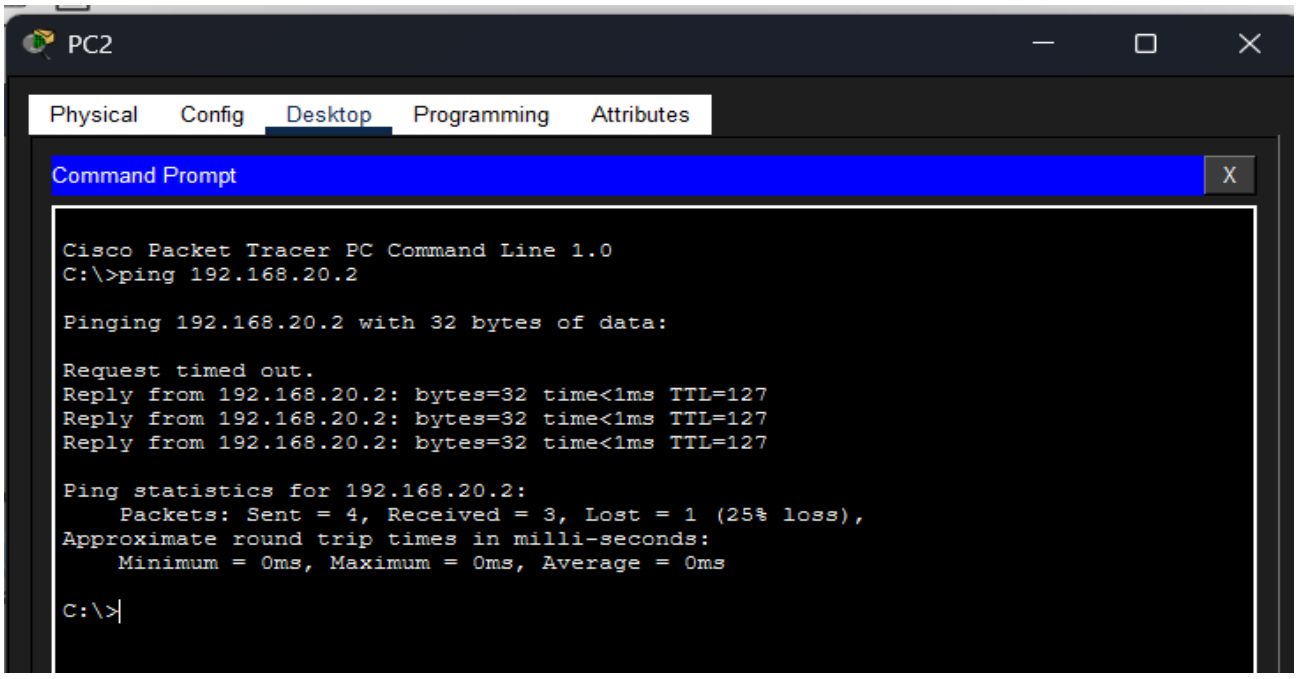


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Task4: 24–30 April

Topic: Inter-VLAN Routing (Basic)





Initializing Flash...
flashfs[0]: 1 files, 0 directories
flashfs[0]: 0 orphaned files, 0 orphaned directories
flashfs[0]: Total bytes: 64016384
flashfs[0]: Bytes used: 4670455

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flashfs[0]: Bytes available: 59345929
flashfs[0]: flashfs fsck took 1 seconds.
...done Initializing Flash.
Boot Sector Filesystem (bs:) installed, fsid: 3
Parameter Block Filesystem (pb:) installed, fsid: 4
Loading "flash:/2960-lanbasek9-mz.150-2.SE4.bin"...
[OK]
Smart Init is enabled
smart init is sizing iomem
TYPE MEMORY_REQ
TOTAL: 0x00000000
Rounded IOMEM up to: 0Mb.
Using 6 percent iomem. [0Mb/512Mb]
Restricted Rights Legend
Use, duplication, or disclosure by the Government is
subject to restrictions as set forth in subparagraph
(c) of the Commercial Computer Software - Restricted
Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
Software clause at DFARS sec. 252.227-7013.
cisco Systems, Inc.
170 West Tasman Drive
San Jose, California 95134-1706
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Initializing flashfs...
fsck: Disable shadow buffering due to heap fragmentation.
flashfs[2]: 2 files, 1 directories
flashfs[2]: 0 orphaned files, 0 orphaned directories
flashfs[2]: Total bytes: 32514048
flashfs[2]: Bytes used: 11952128
flashfs[2]: Bytes available: 20561920
flashfs[2]: flashfs fsck took 2 seconds.
flashfs[2]: Initialization complete....done Initializing flashfs.
Checking for Bootloader upgrade..
Boot Loader upgrade not required (Stage 2)
POST: CPU MIC register Tests : Begin
POST: CPU MIC register Tests : End, Status Passed
POST: PortASIC Memory Tests : Begin
POST: PortASIC Memory Tests : End, Status Passed
POST: CPU MIC interface Loopback Tests : Begin
POST: CPU MIC interface Loopback Tests : End, Status Passed
POST: PortASIC RingLoopback Tests : Begin
POST: PortASIC RingLoopback Tests : End, Status Passed
POST: PortASIC CAM Subsystem Tests : Begin
POST: PortASIC CAM Subsystem Tests : End, Status Passed

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POST: PortASIC Port Loopback Tests : Begin
POST: PortASIC Port Loopback Tests : End, Status Passed
Waiting for Port download...Complete
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
cisco WS-C2960-24TT-L (PowerPC405) processor (revision B0) with 65536K bytes of memory.
Processor board ID FOC1010X104
Last reset from power-on
1 Virtual Ethernet interface
24 FastEthernet interfaces
2 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.
64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address : 00:40:0B:C9:D2:72
Motherboard assembly number : 73-10390-03
Power supply part number : 341-0097-02
Motherboard serial number : FOC10093R12
Power supply serial number : AZS1007032H
Model revision number : B0
Motherboard revision number : B0
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01
Switch Ports Model SW Version SW Image
------ ----- ----- ---------- ----------
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

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%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/24
Switch(config-if)#
Switch con0 is now available
Press RETURN to get started.
Switch>
Switch>enable
Switch#confiure terminal
^
% Invalid input detected at '^' marker.
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

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System Bootstrap, Version 12.1(3r)T2, RELEASE SOFTWARE (fc1)
Copyright (c) 2000 by cisco Systems, Inc.
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Readonly ROMMON initialized
program load complete, entry point: 0x8000f000, size: 0xc940
program load complete, entry point: 0x8000f000, size: 0xc940
program load complete, entry point: 0x8000f000, size: 0x3ed1338
Self decompressing the image :
[OK]
Smart Init is enabled
smart init is sizing iomem
ID TYPE MEMORY_REQ
0003E7 0X00474800 C2811 Mainboard
0X0014B430 Onboard PVDM2 SIMM
0X00264050 Onboard VPN
0X000021B8 Onboard USB
00038E 0X0008BE00 Service Engine AIM
0X002C29F0 public buffer pools
0X00211000 public particle pools

TOTAL: 0X00D85628

If any of the above Memory Requirements are
"UNKNOWN", you may be using an unsupported
configuration or there is a software problem and
system operation may be compromised.
Rounded IOMEM up to: 14Mb.
Using 2 percent iomem. [14Mb/512Mb]
Restricted Rights Legend
Use, duplication, or disclosure by the Government is
subject to restrictions as set forth in subparagraph
(c) of the Commercial Computer Software - Restricted
Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
Software clause at DFARS sec. 252.227-7013.
cisco Systems, Inc.
170 West Tasman Drive
San Jose, California 95134-1706
Cisco IOS Software, 2800 Software (C2800NM-ADVIPSERVICESK9-M), Version 15.1(4)M4
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team
Image text-base: 0x2100F918, data-base: 0x24729040
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you

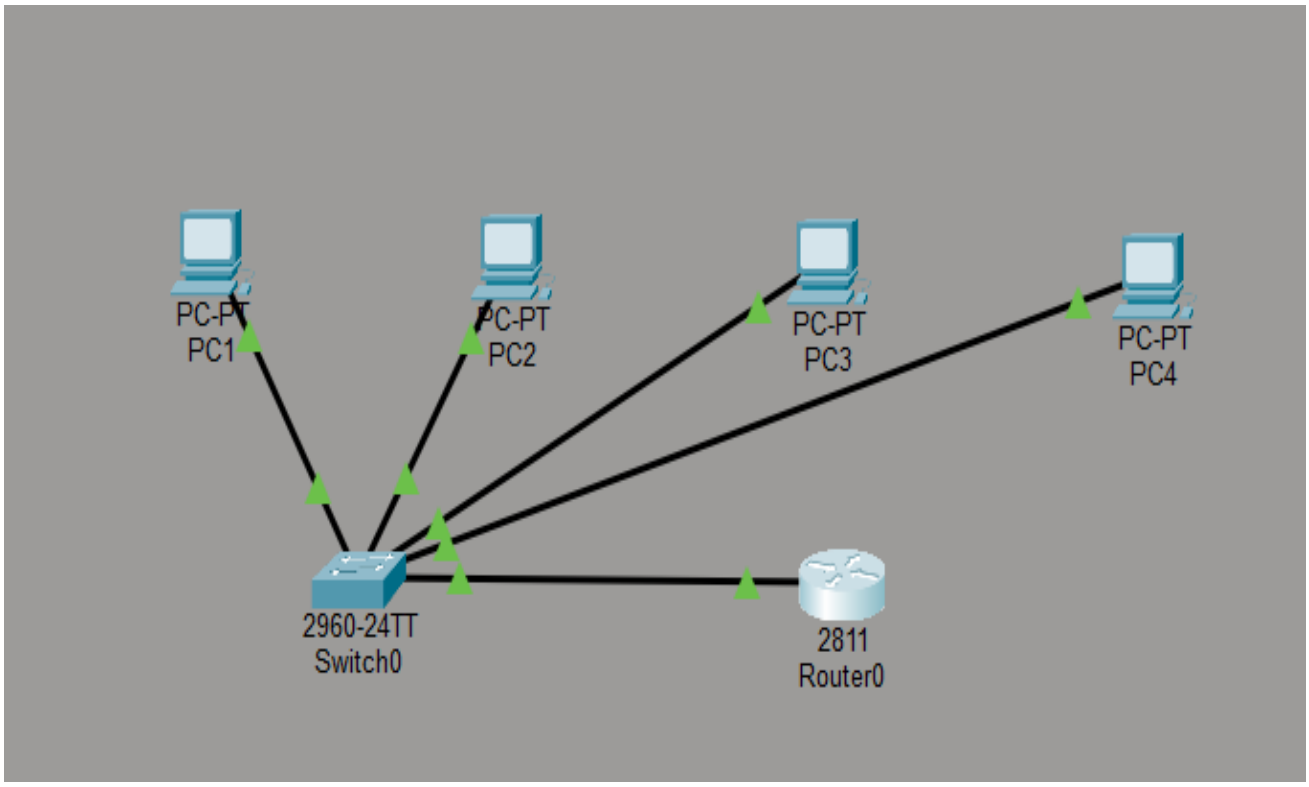
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agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router con0 is now available
Press RETURN to get started.
Press RETURN to get started!
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interfacefa0/0
^
% Invalid input detected at '^' marker.
Router(config)#interface fa0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#interfacefa0/0.10
^
% Invalid input detected at '^' marker.
Router(config)#interface fa0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up
yes
^
% Invalid input detected at '^' marker.
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.254 255.255.255.0

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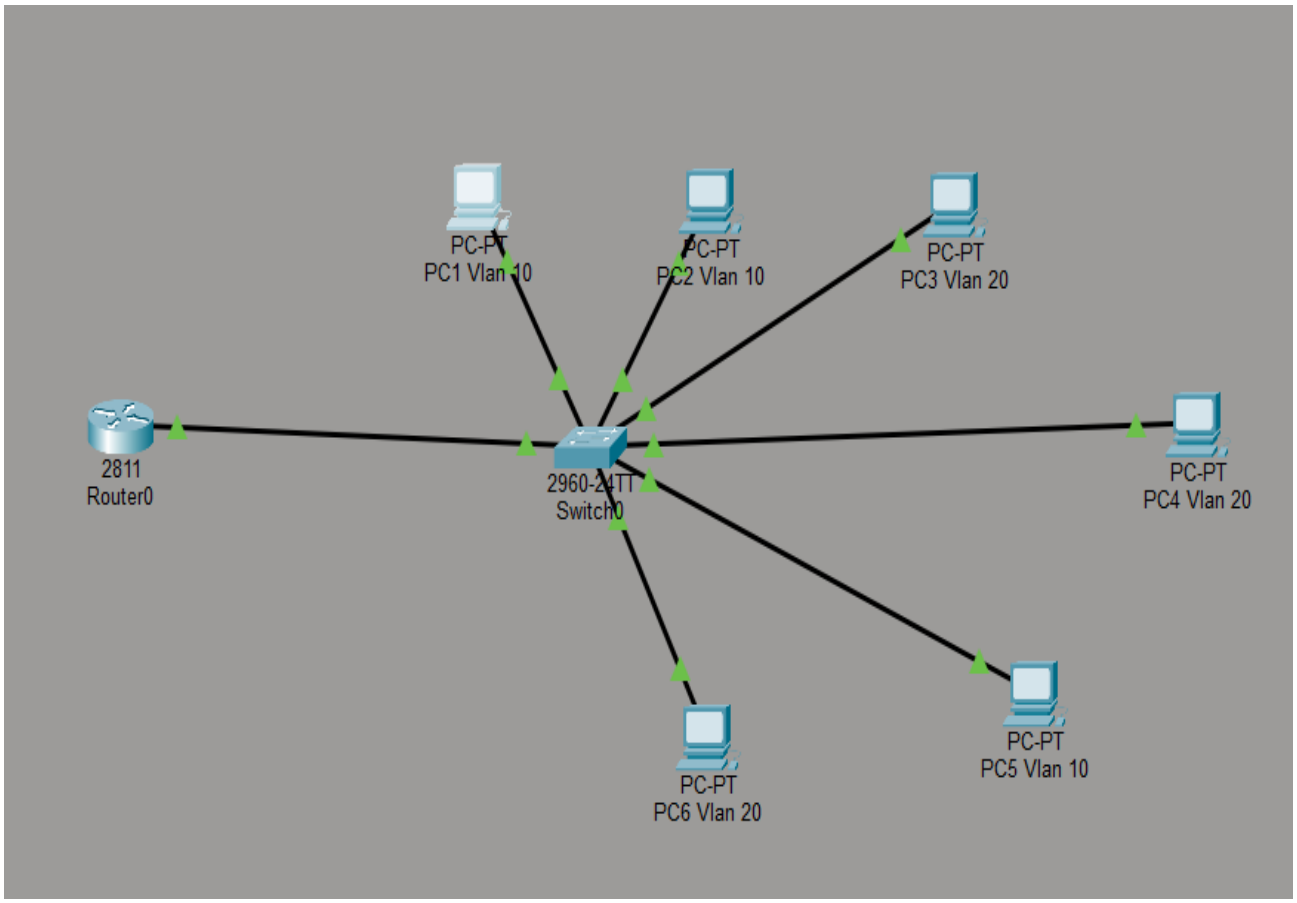
Router(config-subif)#exit
Router(config)#interface fa0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.20, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up
encapulation dot1q 20
^
% Invalid input detected at '^' marker.
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.254 255.255.255.0
Router(config-subif)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Router#
Router con0 is now available
Press RETURN to get started.

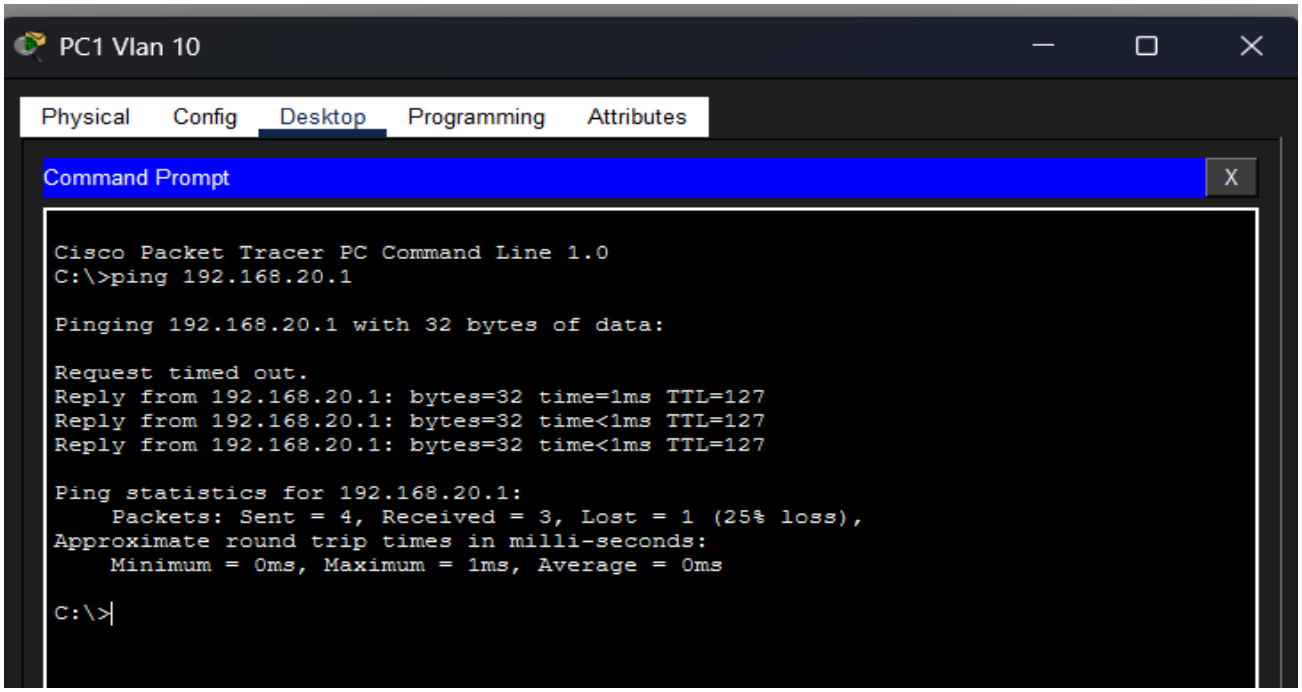


🗓️ Task5: 1–7 May

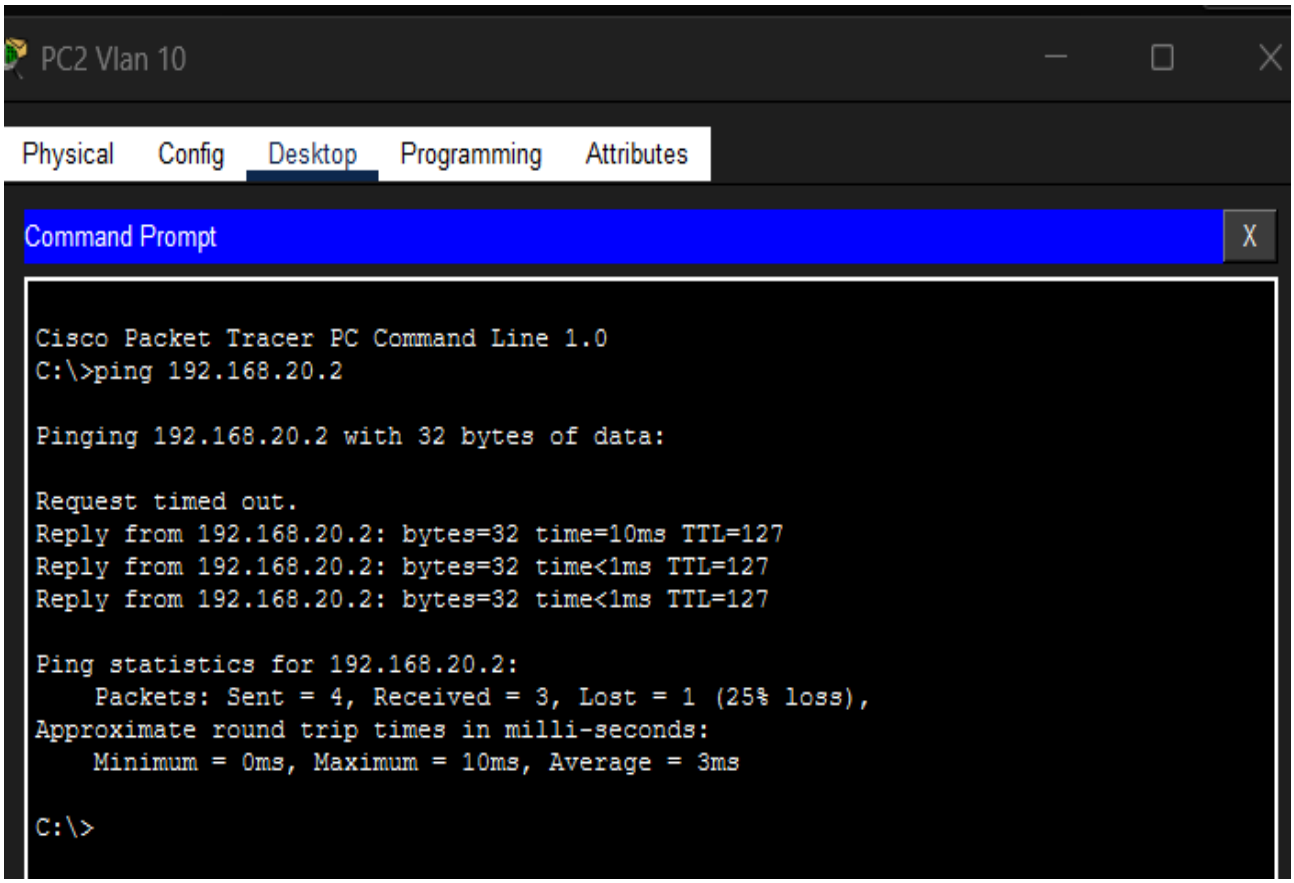
Topic: Final Network Project

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4. Cisco. (2024). *What is Inter-VLAN Routing?* Available at: https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/iproute_bgp/configuration/xe-3s/irb-xe-3s-book.html (Accessed: 25 June 2026).
5. Microsoft. (2022). *Cybersecurity Trends Report*. Microsoft Security Insights.

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